SAEON Agile Logbook

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Field	Record
Role	Backend Developer and Tester
Project	SAEON Real-Time Rainfall Monitoring System
Reporting period	28 July to 23 October 2026
Working days	Tuesday to Friday

Purpose and project configuration

This individual logbook records progress, planned next steps and impediments against the project's 13-week schedule. Weekly headings identify the planned work area; the entries record individual contributions, including supporting work outside a person's main role.

The ESP32 WROOM-32 reads the TR-525M rain gauge. DHT11 is used for testing; SHT31 is the temperature and humidity sensor selected for the final product. Firebase is the database platform.

The agreed architecture retains SIM7000G cellular communication and HiveMQ Cloud MQTT, with a backend ingestion service writing to Firebase. Final deployment and field validation remain open.

Record status

The daily entries are reconstructed working records and require confirmation by the named student. They are not verified attendance or proof that a task was completed. Impediments describe possible risks unless a dated incident is confirmed. Correct any inaccurate activity before signing the logbook.

This revision confirms the technology choices; it does not verify past daily activities or create missing test results. Proposed current follow-up actions appear after the dated entries.

Daily questions

- What did I complete since the last standup?
- What am I planning to complete between now and the next standup?
- What are my impediments (or risks or problems)?

The tables use shortened labels for these same three questions. Mondays are outside the supplied reporting schedule; no Monday attendance or activity is inferred. Historical stand-up meetings are not implied by daily entries.

Week 1 28 July 2026 to 31 July 2026

Planned work area: Project Initiation & System Architecture
PB01 | US01, US02, US03, US04, US13

Tuesday 28 July 2026

Completed since the last stand-up	I reviewed and documented test requirements.
Plan before the next stand-up	I will continue with component test checklist and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Wednesday 29 July 2026

Completed since the last stand-up	I worked on component test checklist.
Plan before the next stand-up	I will continue with development tools and evidence folders and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Thursday 30 July 2026

Completed since the last stand-up	I checked and updated development tools and evidence folders.
Plan before the next stand-up	I will continue with data-flow review and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Friday 31 July 2026

Completed since the last stand-up	I organised the evidence for data-flow review.
Plan before the next stand-up	I will continue with serial output checks and record the results.
Impediments, risks or problems	I need the agreed scope and task ownership to remain clear.

Week 2 4 August 2026 to 7 August 2026

Planned work area: Core Firmware & Sensor Interfacing
PB02 | US01, US02, US11

Tuesday 4 August 2026

Completed since the last stand-up	I reviewed and documented serial output checks.
Plan before the next stand-up	I will continue with temperature reading checks and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Wednesday 5 August 2026

Completed since the last stand-up	I worked on temperature reading checks.
Plan before the next stand-up	I will continue with rain-tip counting tests and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Thursday 6 August 2026

Completed since the last stand-up	I checked and updated rain-tip counting tests.
Plan before the next stand-up	I will continue with firmware and low-power test records and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Friday 7 August 2026

Completed since the last stand-up	I organised the evidence for firmware and low-power test records.
Plan before the next stand-up	I will continue with HiveMQ broker setup and record the results.
Impediments, risks or problems	I depend on working sensors and a stable bench setup.

Week 3 11 August 2026 to 14 August 2026

Planned work area: HiveMQ Cloud Broker Setup
PB03 | US03, US04, US09

Tuesday 11 August 2026

Completed since the last stand-up	I reviewed and documented HiveMQ broker setup.
Plan before the next stand-up	I will continue with topic permissions and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Wednesday 12 August 2026

Completed since the last stand-up	I worked on topic permissions.
Plan before the next stand-up	I will continue with MQTT publish testing and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Thursday 13 August 2026

Completed since the last stand-up	I checked and updated MQTT publish testing.
Plan before the next stand-up	I will continue with Firebase storage integration and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Friday 14 August 2026

Completed since the last stand-up	I organised the evidence for Firebase storage integration.
Plan before the next stand-up	I will continue with modem power test support and record the results.
Impediments, risks or problems	I need consistent message fields, permissions and timestamps.

Week 4 18 August 2026 to 21 August 2026

Planned work area: Cellular Communication (SIM7000G)
PB04 | US03, US04

Tuesday 18 August 2026

Completed since the last stand-up	I reviewed and documented modem power test support.
Plan before the next stand-up	I will continue with SIM registration and APN checks and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Wednesday 19 August 2026

Completed since the last stand-up	I worked on SIM registration and APN checks.
Plan before the next stand-up	I will continue with cellular message validation and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Thursday 20 August 2026

Completed since the last stand-up	I checked and updated cellular message validation.
Plan before the next stand-up	I will continue with reconnection and duplicate handling and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Friday 21 August 2026

Completed since the last stand-up	I organised the evidence for reconnection and duplicate handling.
Plan before the next stand-up	I will continue with dashboard receiver setup and record the results.
Impediments, risks or problems	I depend on cellular coverage, an active SIM and stable modem power.

Week 5 25 August 2026 to 28 August 2026

Planned work area: SAEON Dashboard Base Application
PB05 | US05, US06, US09

Tuesday 25 August 2026

Completed since the last stand-up	I reviewed and documented dashboard receiver setup.
Plan before the next stand-up	I will continue with live-data message checks and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Wednesday 26 August 2026

Completed since the last stand-up	I worked on live-data message checks.
Plan before the next stand-up	I will continue with historical-data query checks and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Thursday 27 August 2026

Completed since the last stand-up	I checked and updated historical-data query checks.
Plan before the next stand-up	I will continue with cloud-command interface preparation and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Friday 28 August 2026

Completed since the last stand-up	I organised the evidence for cloud-command interface preparation.
Plan before the next stand-up	I will continue with battery test records and record the results.
Impediments, risks or problems	I need the dashboard to handle missing and outdated data clearly.

Evidence reference and entry corrections: _____

Week 6 1 September 2026 to 4 September 2026

Planned work area: Power Supply & Solar Integration
PB06 | US11

Entry status: reconstructed; student confirmation required.

Tuesday 1 September 2026

Completed since the last stand-up	I reviewed and documented battery test records.
Plan before the next stand-up	I will continue with solar test support and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Wednesday 2 September 2026

Completed since the last stand-up	I worked on solar test support.
Plan before the next stand-up	I will continue with mains-power alternative testing and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Thursday 3 September 2026

Completed since the last stand-up	I checked and updated mains-power alternative testing.
Plan before the next stand-up	I will continue with power-flow audit and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Friday 4 September 2026

Completed since the last stand-up	I organised the evidence for power-flow audit.
Plan before the next stand-up	I will continue with station-health data validation and record the results.
Impediments, risks or problems	I need verified power records before claiming an operating duration.

Week 7 8 September 2026 to 11 September 2026

Planned work area: Advanced UI & Health Diagnostics
PB07 | US07, US08, US10

Tuesday 8 September 2026

Completed since the last stand-up	I reviewed and documented station-health data validation.
Plan before the next stand-up	I will continue with CSV export functionality and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Wednesday 9 September 2026

Completed since the last stand-up	I worked on CSV export functionality.
Plan before the next stand-up	I will continue with responsive-data checks and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Thursday 10 September 2026

Completed since the last stand-up	I checked and updated responsive-data checks.
Plan before the next stand-up	I will continue with end-to-end pipeline testing and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Friday 11 September 2026

Completed since the last stand-up	I organised the evidence for end-to-end pipeline testing.
Plan before the next stand-up	I will continue with circuit continuity test planning and record the results.
Impediments, risks or problems	I need diagnostic and exported values to match the stored readings.

Week 8 15 September 2026 to 18 September 2026

Planned work area: Hardware Finalization & PCB Soldering
PB08 | US12

Entry status: reconstructed; student confirmation required.

Tuesday 15 September 2026

Completed since the last stand-up	I reviewed and documented circuit continuity test planning.
Plan before the next stand-up	I will continue with soldering inspection support and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Wednesday 16 September 2026

Completed since the last stand-up	I worked on soldering inspection support.
Plan before the next stand-up	I will continue with enclosure wiring checks and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Thursday 17 September 2026

Completed since the last stand-up	I checked and updated enclosure wiring checks.
Plan before the next stand-up	I will continue with assembled hardware bench checks and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Friday 18 September 2026

Completed since the last stand-up	I organised the evidence for assembled hardware bench checks.
Plan before the next stand-up	I will continue with power-drain measurements and record the results.
Impediments, risks or problems	I need the permanent assembly to match the reviewed wiring.

Evidence reference and entry corrections: _____

Week 9 22 September 2026 to 25 September 2026

Planned work area: Pre-Deployment Auditing & Calibration
PB09 | US11, US12

Entry status: reconstructed; student confirmation required.

Tuesday 22 September 2026

Completed since the last stand-up	I reviewed and documented power-drain measurements.
Plan before the next stand-up	I will continue with solar-charging test evidence and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Wednesday 23 September 2026

Completed since the last stand-up	I worked on solar-charging test evidence.
Plan before the next stand-up	I will continue with installation test preparations and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Thursday 24 September 2026

Completed since the last stand-up	I prepared notes on installation test preparations. The field work remains in progress.
Plan before the next stand-up	I will continue with field-calibration test preparations and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Friday 25 September 2026

Completed since the last stand-up	I prepared notes on field-calibration test preparations. The field work remains in progress.
Plan before the next stand-up	I will continue with field-test logging setup and record the results.
Impediments, risks or problems	I still depend on completion of installation and baseline field calibration.

Evidence reference and entry corrections: _____

Week 10 29 September 2026 to 2 October 2026

Planned work area: 48-Hour Field Stress Test
PB10 | US01, US02, US03, US11, US12

Entry status: reconstructed; student confirmation required.

Tuesday 29 September 2026

Completed since the last stand-up	I prepared notes on field-test logging setup. The field work remains in progress.
Plan before the next stand-up	I will continue with available monitoring records and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Wednesday 30 September 2026

Completed since the last stand-up	I prepared notes on available monitoring records. The field work remains in progress.
Plan before the next stand-up	I will continue with reference-data comparison preparation and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Thursday 1 October 2026

Completed since the last stand-up	I prepared notes on reference-data comparison preparation. The field work remains in progress.
Plan before the next stand-up	I will continue with provisional field-audit notes and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Friday 2 October 2026

Completed since the last stand-up	I prepared notes on provisional field-audit notes. The field work remains in progress.
Plan before the next stand-up	I will continue with test findings for slides and record the results.
Impediments, risks or problems	I cannot treat field validation as complete while the test and audit remain in progress.

Evidence reference and entry corrections: _____

Week 11 6 October 2026 to 9 October 2026

Planned work area: Presentation Formatting & Content
PB11 | US13

Entry status: reconstructed; student confirmation required.

Tuesday 6 October 2026

Completed since the last stand-up	I reviewed and documented test findings for slides.
Plan before the next stand-up	I will continue with data-flow diagram checks and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Wednesday 7 October 2026

Completed since the last stand-up	I worked on data-flow diagram checks.
Plan before the next stand-up	I will continue with dashboard evidence validation and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Thursday 8 October 2026

Completed since the last stand-up	I checked and updated dashboard evidence validation.
Plan before the next stand-up	I will continue with backend setup documentation and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Friday 9 October 2026

Completed since the last stand-up	I organised the evidence for backend setup documentation.
Plan before the next stand-up	I will continue with technical slide checks and record the results.
Impediments, risks or problems	I need the presentation to distinguish prototype results from pending field results.

Evidence reference and entry corrections: _____

Week 12 13 October 2026 to 16 October 2026

Planned work area: Rehearsal & Submission Prep
PB12 | US13

Entry status: reconstructed; student confirmation required.

Tuesday 13 October 2026

Completed since the last stand-up	I reviewed and documented technical slide checks.
Plan before the next stand-up	I will continue with demo pipeline rehearsal and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Wednesday 14 October 2026

Completed since the last stand-up	I worked on demo pipeline rehearsal.
Plan before the next stand-up	I will continue with rehearsal fault checks and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Thursday 15 October 2026

Completed since the last stand-up	I checked and updated rehearsal fault checks.
Plan before the next stand-up	I will continue with prototype build and evidence package and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Friday 16 October 2026

Completed since the last stand-up	I organised the evidence for prototype build and evidence package.
Plan before the next stand-up	I will continue with prototype release review and record the results.
Impediments, risks or problems	I need a reliable demonstration and clear documentation of open field work.

Evidence reference and entry corrections: _____

Week 13 20 October 2026 to 23 October 2026

Planned work area: Project Buffer & Final Review
PB13 | US12, US13

Entry status: reconstructed; student confirmation required.

Tuesday 20 October 2026

Completed since the last stand-up	I reviewed and documented prototype release review.
Plan before the next stand-up	I will continue with test-evidence links and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Wednesday 21 October 2026

Completed since the last stand-up	I worked on test-evidence links.
Plan before the next stand-up	I will continue with remaining field-validation requirements and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Thursday 22 October 2026

Completed since the last stand-up	I checked and updated remaining field-validation requirements.
Plan before the next stand-up	I will continue with prototype configuration record and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Friday 23 October 2026

Completed since the last stand-up	I organised the evidence for prototype configuration record.
Plan before the next stand-up	I will continue with the remaining deployment and field-validation actions and record the results.
Impediments, risks or problems	I am still awaiting completion of final deployment and field validation.

Evidence reference and entry corrections: _____

Taskboard summary and current follow-up

Snapshot retained from the project backlog at 23 October 2026; document follow-up added on 27 October 2026. These shared statuses do not mean that this student personally completed every task.

Product Backlog	In Progress	Completed
PB17: Final SHT31 verification. PB18: Firebase access and ingestion checks. PB19: Assessment evidence package.	9.03 Installation. 9.04 Baseline calibration. 10.01 Test initiation. 10.02 Active monitoring. 10.03 Reference measurements. 10.04 Audit and conclusions.	43 original task cards reported complete. This covers planning, prototype development and bench work, documentation and presentation preparation. Evidence checks remain necessary.
Optional: PB14 pressure, wind and UV; PB15 GPS; PB16 over-the-air updates.	Field acceptance remains open.	Prototype documentation sign-off is separate from field acceptance.

Taskboard walkthrough

Current review on 27 October 2026. These responses record the present follow-up actions, not meetings backdated to the reporting period.

What do we need to do to advance this piece of work

Confirm the SHT31 final configuration and Firebase evidence, finish installation and baseline calibration, then complete and audit the 48-hour field test.

Is anyone working on anything that is not on the board

Ethan’s 4–7 August entries include simulation work, HTTP connection investigation and a supervisor meeting about a pre-seed presentation. Cross-reference these as historical support activities; do not replace them with planned sprint tasks. Record any further unlisted work when confirmed.

What do we need to finish as a team

Close the six open field tasks, confirm PB17–PB19 evidence, verify individual daily entries and complete the graduate-attribute and GitHub portfolio evidence.

Are there any bottlenecks or blockers to the flow of work

Open dependencies include installation, calibration, final-sensor evidence and field-test records. Site access, modem power and connectivity are risks to check; do not record them as incidents without dated evidence.

My next action

Verify Firebase ingestion and access rules, and collect the connection, power and field-test evidence.

Verification and portfolio evidence

Before submission, the named student should confirm the dated entries, replace unsupported statements, identify actual risks or incidents and attach the relevant evidence. Keep any correction transparent. Do not insert invented work hours, signatures or meetings.

Check	What to record
Identity	Complete any missing student number.
Daily activity	Confirm what was done, the next planned action and actual impediments for each date.
Evidence	Link the relevant code, notes, screenshots, test records or meeting records.
Final sensor	Identify DHT11 testing and SHT31 final-product evidence clearly.
Graduate attributes	Personalise the companion reflections and link genuine examples of the four attributes.
Reporting period	Confirm the accepted dates and Tuesday-to-Friday arrangement with the lecturer.
Review	Record the reviewer, date and corrections after the review occurs.

Student confirmation

I have checked the dated entries against my own work and corrected any inaccurate or unsupported statements. My evidence references identify the work that I contributed.

Name: Qaasim Isaacs
Signature: _____ Date: _____

Reviewer record

Reviewer name: _____
Review date: _____
Comments and required corrections: _____

Reporting period and references

This records the supplied project schedule. The subject guide lists March to June for the eWIL assessment activities and July to November for WIL. The applicable assessment deadline and acceptance of this reporting period must be confirmed with the lecturer; the dates here are not a revised university deadline.

Requirement reference: PRP37XS Subject Guide, pages 20–21 for the taskboard, walkthrough and daily questions; pages 24–25 for graduate attributes. Related documents: Team Charter, Product Backlog, Sprint Backlog, User Stories and Tasks, and Graduate Attributes and Portfolio Evidence.